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Evaluating the effects of ICT engagement on science and mathematics achievement: Evidence from PISA 2022 using structural equation modeling

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A B S T R A C T

PISA-based research has often concluded that students' ICT use is weakly, inconsistently, or negatively related to academic achievement, reinforcing the assumption that technology investments deliver limited learning benefits. This study challenges that conclusion by arguing that such findings stem from treating ICT engagement as a homogeneous and linear exposure rather than as a differentiated, psychologically mediated learning condition. Grounded in social cognitive, cognitive load, and engagement theories, we conceptualize ICT effects as conditional on both the mode and intensity of use and students' digital self-efficacy. Analyzing PISA 2022 data from Türkiye, an emerging-economy context with large-scale ICT reforms, it is shown that ICT engagement exhibits systematically different relationships with mathematics and science achievement. Academically oriented ICT use outside the classroom is positively associated with achievement, whereas leisure-oriented ICT use is negatively associated, demonstrating that increased ICT exposure alone does not improve learning. Moreover, these relationships are non-linear. Moderate, purposeful ICT use is beneficial, but higher levels yield diminishing or adverse returns, offering a theoretical explanation for the mixed results reported in prior PISA studies. Crucially, digital self-efficacy mediates key ICT–achievement pathways, indicating that ICT contributes to learning primarily by strengthening students' confidence and capacity to use digital tools strategically. By integrating differentiated ICT use, non-linear effects, and psychological mediation within a single explanatory framework, this study advances PISA-based ICT research beyond access- and usage-frequency models. The findings reframe ICT not as a general educational input but as a pedagogically contingent resource whose effectiveness depends on learner readiness and instructional purpose, with direct implications for digital education policy in expanding systems.

1. Introduction

The rapid digitalization of education has positioned Information and Communication Technology (ICT) as a central instrument in efforts to improve student learning, particularly in foundational fields such as mathematics and science. Across education systems, large-scale investments in devices, connectivity, and digital platforms have been justified by the expectation that ICT-enhanced instruction will foster engagement, deepen conceptual understanding, and ultimately increase achievement (OECD, 2023a; Wang & Wang, 2023). These expectations have been especially prominent in emerging-economy contexts, including Türkiye, where national initiatives such as the FATIH project have aimed to expand digital access and modernize instructional practices (Kaya & Yılayaz, 2013; Ünal et al., 2022). Yet despite sustained policy attention and financial commitment, the empirical evidence linking ICT engagement to academic achievement remains equivocal.

Prior research documents a persistent inconsistency in ICT–achievement findings. Meta-analyses and quasi-experimental studies indicate that ICT can support learning when integrated purposefully and aligned with pedagogical goals (Tamim et al., 2011; Reguera & López, 2021). In contrast, large-scale observational studies particularly those drawing on PISA data frequently report weak, null, or negative associations between ICT use and achievement, especially when ICT engagement is intensive, unguided, or oriented toward leisure activities (Bulut & Cutumisu, 2018; Masood et al., 2020; Zhu & Li, 2022). This divergence presents a substantive challenge for both

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researchers and policymakers. While technology is increasingly embedded in schooling, its educational value remains theoretically under-specified and empirically contested.

A key source of this inconsistency lies in how ICT engagement has typically been conceptualized. Much of the literature operationalizes ICT use as a single, aggregated construct often measured by frequency or access implicitly assuming that more ICT use should correspond to greater learning gains. Such approaches overlook the multidimensional nature of ICT engagement and obscure important distinctions between qualitatively different forms of use, such as ICT use at school, ICT use for academic purposes outside the classroom, and leisure-oriented ICT use (Skryabin et al., 2015; Zhu & Li, 2022). Treating these modes as interchangeable risks conflating pedagogically productive engagement with forms of ICT use that may distract from learning or impose additional cognitive demands.

A second limitation concerns the functional form of ICT–achievement relationships. A growing body of evidence suggests that ICT effects are not strictly linear. Rather, moderate and structured ICT use may enhance learning, whereas excessive or poorly designed ICT engagement can lead to diminishing or even negative returns (Li & Zhu, 2023; Ünal et al., 2022). These findings align with Cognitive Load Theory, which posits that learning is optimized when instructional supports enhance germane cognitive processing without overloading working memory (Sweller, 1988). However, many large-scale studies continue to rely on linear specifications, thereby masking threshold effects that could reconcile contradictory findings in literature.

Equally important is the limited attention given to psychological mechanisms that mediate ICT effects. Educational psychology emphasizes that technology influences learning indirectly, through students' motivational beliefs, self-regulatory behaviors, and perceptions of competence. Digital self-efficacy, grounded in Social Cognitive Theory, has been identified as a particularly prominent mechanism shaping how students engage with and benefit from ICT (Bandura, 1997; Hatlevik et al., 2018). Students who lack confidence in their digital abilities may use technology superficially or avoid cognitively demanding tasks, whereas students with higher self-efficacy are more likely to engage strategically and persist in learning activities (Zimmerman, 2000). Despite this theoretical importance, few PISA-based studies explicitly model digital self-efficacy as a mediator linking ICT engagement to achievement.

Recent developments have created an opportunity to address these limitations more systematically. The PISA 2022 cycle includes expanded ICT modules capturing multiple dimensions of ICT engagement, alongside self-reported measures of digital competence, collected in a post-pandemic context characterized by intensified and diversified technology use (OECD, 2023a; OECD, 2023b). These data enable a more fine-grained analysis of ICT engagement than was possible in earlier cycles. At the same time, advances in structural equation modeling facilitate the simultaneous estimation of latent constructs, mediation pathways, and non-linear relationships while accounting for measurement error and plausible values (Hair et al., 2019; Rutkowski et al., 2010).

Against this backdrop, the present study examines how differentiated modes of ICT engagement relate to mathematics and science achievement using PISA 2022 data from Türkiye. Türkiye represents a theoretically informative case due to its extensive ICT investments, heterogeneous implementation contexts, and prior evidence of mixed ICT–achievement relationships (Yılmaz Koğar & Koğar, 2019; Ünal et al., 2022). Rather than asking whether ICT use is beneficial in general, this study investigates under what conditions ICT engagement supports learning, focusing on usage mode, intensity, and students' digital self-efficacy.

The study integrates insights from Social Cognitive Theory (Bandura, 1997), Self-Regulated Learning Theory (Zimmerman, 2000), Engagement Theory (Kearsley & Shneiderman, 1998), and Cognitive Load Theory (Sweller, 1988) to advance a conditional model of ICT effectiveness. This model posits that ICT contributes to achievement only when engagement is pedagogically aligned, cognitively manageable, and supported by students' confidence in their digital capabilities. Using latent-variable structural equation modeling, the study estimates both linear and non-linear effects of multiple ICT engagement modes and evaluates digital self-efficacy as a mediating mechanism.

By addressing conceptual oversimplification, functional form misspecification, and missing psychological mechanisms, this study responds directly to key limitations in prior ICT–achievement research. It contributes theoretically by offering an integrated explanatory framework, empirically by providing post-pandemic, country-level evidence from Türkiye, and methodologically by demonstrating the value of modeling mediation and non-linearity in large-scale assessment data. In doing so, it advances a more precise and analytically grounded understanding of ICT's role in mathematics and science achievement within and beyond the PISA context.

In summary, prior PISA-based work often leaves unresolved why ICT–achievement findings are so mixed because it treats ICT as a single, frequency-based exposure and uses primarily linear specifications without embedding psychological mechanisms; as a result, aggregate analyses conflate pedagogically productive uses with distracting or cognitively costly uses and cannot explain threshold or context-dependent effects. This study addresses that exact shortcoming by (a) disaggregating ICT engagement into in-school, at-home academic, out-of-class school-task, and leisure modes, (b) explicitly testing non-linear (quadratic) functional forms to detect diminishing or adverse returns, and (c) integrating digital self-efficacy as a theoretically motivated mediator within a latent-variable SEM framework applied to PISA 2022 Türkiye data thereby moving beyond access/frequency accounts to show when and through what ICT use supports mathematics and science learning. This combined conceptual and empirical strategy directly resolves the three core limitations that leave prior PISA studies equivocal (homogenized measurement, mis-specified functional form, and omitted psychological mediation) and produces a falsifiable, policy-relevant account of ICT effectiveness in a post-pandemic, emerging-economy context.

The purpose of this study is to examine the linear and non-linear relationships between students' ICT use at school, ICT use for outside-of-school learning, ICT use for leisure, and using ICT as students' academic achievement in mathematics and science in PISA 2022.

Research questions guiding the study are:

1. To what extent are ICT use in school and ICT use for academic activities outside school linearly related to digital self-efficacy and to mathematics and science achievement?
2. To what extent does digital self-efficacy mediate the relationship between ICT use in school and ICT use for academic activities outside school, and mathematics and science achievement?
3. What is the nature of the non-linear relationships among ICT use in school, ICT use for academic activities outside school, digital self-efficacy, and mathematics and science achievement?

Answering these questions advances evidence-based policy by clarifying when ICT matters for STEM learning and what conditions maximize its returns.

2. Literature review

The rapid digital transformation of societies has placed growing demands on education systems to equip students with the skills and mindsets needed for a world in which Information and Communication Technology (ICT) enters nearly every aspect of learning, work, and daily life. Thus, developing ICT competence is not only essential for future employability but also holds promise for enriching current instruction in foundational areas such as mathematics and science (Budiarto et al., 2024; Ma, et al., 2024). Although substantial investments in ICT infrastructure and policy initiatives have been made globally, empirical findings on how various forms of ICT engagement relate to student outcomes remain inconclusive. Some studies report positive effects under purposeful, pedagogically grounded integration, whereas others observe negligible or even adverse effects when ICT use is unguided, excessive, or misaligned with learner needs and these patterns appear contingent on learner attributes and broader contextual factors (Tamim et al., 2011; Lei & Zhao, 2007; Skryabin et al., 2015; Ünal et al., 2022).

Moreover, gaps persist in understanding these dynamics within emerging-economy settings, where infrastructure quality, pedagogical practices, and student ICT readiness vary widely. To address these complexities and inform both theory and practice, the present study first outlines distinct categories of student ICT use such as in-class integration, out-of-class academic tasks, home-based learning activities, leisure-oriented use, and exploratory, then examines theoretical pathways by drawing on engagement theory, self-regulated learning, cognitive load, and self-efficacy frameworks that may link each category to mathematics and science achievement. Digital self-efficacy is introduced early as a potential mediator shaping how students utilize ICT for learning, and study goals, hypotheses, and a conceptual model are articulated by defining all variables to guide rigorous analysis of linear and non-linear relationships using structural equation modeling.

Several factors motivate this study. First, the global shift toward digitalized instruction and blended learning modalities accelerated by recent events makes understanding ICT's educational impact imperative (Budiarto et al., 2024; Odell et al., 2020). Second, while some studies report positive associations between purposeful ICT integration and achievement (Tamim et al., 2011; Lei & Zhao, 2007), others find null or negative effects when ICT use is unguided or excessive (Skryabin et al., 2015; Bulut & Cutumisu, 2018; Zhu & Li, 2022). Third, the majority of existing research focuses on western or high-income countries. There is a lack of detailed studies in emerging-economy settings, where ICT infrastructure, teaching methods, and student backgrounds vary (Yılmaz Koğar & Koğar, 2019; Turgut & Aslan, 2021). Therefore, examining Türkiye that has a large, diverse emerging-economy context with notable ICT integration initiatives such the FATIH project can yield insights relevant to many countries undergoing similar transitions (Kaya & Yılayaz, 2013; Ünal et al., 2022). Moreover, PISA comparisons indicate that ICT–achievement relationships vary cross-nationally (Courtney et al., 2022), highlighting the value of context-specific studies to inform global theory and policy.

Empirical studies and large-scale assessments (Hatlevik et al., 2018) distinguish multiple ICT engagement modes. In-school ICT use refers to digital tools integrated during lesson time such as simulations, interactive platforms; ICT use for school-related tasks outside class encompasses online communication with teachers, digital submission of assignments, and access to instructional resources; home ICT use for learning involves engaging with educational platforms or resources after school; leisure ICT use refers to non-academic screen activities, including gaming and the use of social media, that are not directly related to school learning tasks.

Multiple theoretical perspectives are drawn on to explain how different ICT engagement modes may influence mathematics and science learning. Engagement Theory posits that interactive, meaningful ICT tasks foster cognitive engagement and deeper understanding (Kearsley & Shneiderman, 1998); Self-Regulated Learning Theory highlights that ICT tools supporting planning, monitoring, and reflection can scaffold autonomous study behaviors and thus enhance achievement (Zimmerman, 2000); Cognitive Load Theory warns that poorly designed or excessive ICT activities impose extraneous cognitive load that impedes learning, emphasizing the need for optimal instructional design and moderation (Sweller, 1988); Self-Determination Theory illuminates how satisfying basic psychological needs for autonomy, competence, and relatedness in ICT contexts shapes cognitive-motivational engagement such as perceived autonomy in ICT use which may mediate links between ICT engagement and achievement (Ryan & Deci, 2017; Li & Zhu, 2023; Bhutoria & Aljabri, 2022); and Self-Efficacy Theory emphasizes that learners' confidence in their ability to perform ICT-related tasks influences whether they engage productively, with higher digital self-efficacy correlating with more strategic ICT use and better learning outcomes (Bandura, 1997; Hatlevik et al., 2018; Skryabin et al., 2015).

Studies have modeled linear ICT and achievement associations, often finding weak or context-dependent effects (Lei & Zhao, 2007; Skryabin et al., 2015). However, more recent research has uncovered non-linear patterns, indicating that moderate, structured ICT use can benefit learning up to a threshold beyond which returns diminish or even reverse (Zhu & Li, 2022; Li & Zhu, 2023; Ünal et al., 2022), making it essential to capture these dynamics rather than assume that increased ICT use is always better. Structural equation modeling (SEM) is particularly advantageous for this purpose because it allows the modeling of latent constructs such as digital self-efficacy, cognitive-motivational engagement, accounts for measurement error, and simultaneously tests complex mediation such as self-efficacy or self-determination theory–based dispositions and moderation effects such as socioeconomic status, pedagogical quality, including non-linear terms, while providing overall model fit evaluation (Kline, 2016; Hair et al., 2010). Indeed, several recent studies have employed SEM to examine ICT–achievement pathways (Srijamdee & Pholphirul, 2020; Li & Zhu, 2023; Odell et al., 2020). Although multilevel and cross-national analyses using hierarchical models have explored student, school, and country level factors often finding that attitudes and dispositions such as perceived autonomy show stronger links to achievement than mere ICT availability (Courtney et al., 2022; Rehman, Huang, Mahmood, 2025). SEM offers unique strengths in modeling latent mediators and testing theory-driven pathways within and across contexts.

By integrating these strands, it has been identified that: (a) ICT engagement modes likely have distinct effects on learning via different mechanisms; (b) learner dispositions (self-efficacy, SDT-based engagement) play crucial mediating roles; (c) non-linear patterns must be examined to detect optimal engagement levels; (d) SEM is well-suited for testing these complex, theory-driven models; and (e) context matters. Findings from one country may not generalize, so studying Türkiye contributes valuable evidence.

Türkiye presents a compelling case for international readers as a large, diverse emerging-economy with concerted ICT-integration reforms, it exemplifies contexts where infrastructure, pedagogical practices, and student ICT competencies vary widely (Kaya & Yılayaz, 2013; Turgut & Aslan, 2021). Prior PISA-based analyses in Türkiye reveal moderate-to-low ICT use in schools and predominant out-of-school ICT engagement (Yılmaz Koğar & Koğar, 2019; Ünal et al., 2022). Understanding how different ICT use modes relate to mathematics and science achievement in Türkiye can inform policy both locally and in similar contexts globally, offering insights into boundary conditions and effective practices. Moreover, comparative studies emphasize cross-national variability in ICT–learning relationships (Courtney et al., 2022), highlighting the necessity of context-specific SEM analyses to refine global theories.

The following details outline the relationships between ICT and students' academic performance.

2.1. ICT use and academic achievement

ICT use in education has been linked to improved academic achievement (Zhu & Li, 2022). Studies have shown that integrating ICT tools like interactive whiteboards and educational software enhances student engagement, motivation, and performance (Reguera & Lopez, 2021; Remón et al., 2017; Sun & Hsieh, 2018; Alisoy, 2023). Access to online resources and self-directed learning through ICT devices has also been associated with deeper understanding and improved academic success (Corchuelo-Fernández et al., 2022; Onah et al., 2021; Sari, 2022). Overall, embracing ICT in education creates dynamic learning environments that promote academic excellence (Abrenilla et al.,2023).

2.2. ICT available at school and academic achievement

The availability of ICT in schools has a significant impact on academic achievement. When schools provide access to ICT tools such as computers, internet connectivity, and educational software, students benefit from enhanced learning opportunities. Research shows that students who have access to ICT resources at school demonstrate improved academic performance (Hussain et al., 2017; Wang & Wang, 2023; Valverde-Berrocoso et al., 2022; Timotheou et al., 2023). Access to ICT enables students to engage in interactive learning experiences, access a wealth of information, and develop digital literacy skills. This integration of ICT in schools has the potential to positively influence students' academic achievements and prepare them for success in the digital age.

2.3. ICT available outside school and academic achievement

In today's digital age, ICT has become an integral part of our daily lives. With the rapid advancement of technology, ICT tools are increasingly accessible outside the traditional educational setting. There are some potential benefits of ICT in facilitating learning beyond the classroom. One of the key advantages of ICT availability outside school is the easy access to a vast amount of information. Digital platforms, such as the internet and online libraries, provide students with a plethora of resources to enhance their knowledge and understanding. Research has indicated that increased access to information through ICT positively correlates with improved academic performance (Chen et al., 2024; Miraj et al., 2021; Lee & Wu, 2012). Students can utilize search engines, digital textbooks, and educational websites to explore topics beyond the confines of traditional classroom materials, leading to a more comprehensive understanding of subjects.

In addition, ICT tools offer the potential for personalized learning experiences outside the school environment. Adaptive learning platforms and intelligent tutoring systems can tailor educational content to individual students' needs, allowing them to progress at their own pace. Studies have shown that personalized learning through ICT can lead to enhanced student engagement, motivation, and academic achievement (Tsai et al., 2020; Hwang et al., 2015; Montebello, 2021). With the ability to access educational resources tailored to their specific learning styles and preferences, students can develop a deeper understanding of the material and achieve better academic outcomes.

The third advantage of ICT availability outside school also facilitates collaborative learning opportunities. Online platforms, discussion forums, and video conferencing tools enable students to interact with peers, educators, and experts from around the world. Collaborative learning has been linked to improved academic performance, as it encourages active participation, knowledge sharing, and critical thinking (Kearney et al., 2012). Through ICT-mediated collaboration, students can engage in virtual group projects, exchange ideas, and receive feedback, thereby enhancing their academic achievement.

Lastly, the use of ICT outside school provides students with valuable digital skills that are increasingly essential in today's workforce. Proficiency in digital literacy, information management, and online communication are crucial for success in higher education and future careers. Research has shown a positive relationship between ICT usage outside school and the development of digital skills, which, in turn, contribute to improved academic performance (Livingstone et al., 2017; Limniou et al., 2021; Al-Rahmi et al., 2020; Hu et al., 2018). By utilizing ICT tools for research, content creation, and communication, students can acquire and refine the skills necessary for academic success.

2.4. ICT use for outside-of-school learning and academic achievement

Outside of formal educational settings, learning can pose more challenges compared to learning within the structured environment of schools, where support from teachers is readily available (Singal & Swann, 2009). Introducing ICT into outside-of-school learning endeavors can serve as a scaffold for students' educational journeys. ICT tools facilitate communication between students and teachers regarding schoolwork and enable students to extend their learning through internet resources or educational apps. This integration of technology may aid students in overcoming learning obstacles by providing additional avenues for clarification and practice. However, the incorporation of ICT into learning experiences can also introduce complexities. Students engaging with ICT may experience an increased cognitive load as they navigate both the real-world learning tasks and the digital environment within a limited timeframe (Skulmowski & Xu, 2022). This dual focus may lead to a diffusion of mental effort, potentially detracting from the depth of engagement with the learning material (Eryilmaz et al., 2009), consequently impacting learning outcomes negatively.

In summary, while ICT holds promise in enhancing outside-of-school learning by providing additional support and resources, its introduction also demands careful consideration of its potential impact on cognitive load and learning efficacy.

2.5. Leisure activities using ICT and academic achievement

Numerous studies have explored the relationship between leisure activities that involve the use of ICT and academic achievement. Research has consistently shown that engaging in leisure activities that incorporate ICT can have a positive impact on academic performance (Ben Youssef et al., 2022; Mokhtarian et al., 2006; Courtney et al., 2022). For example, adolescents who spent more time on educational ICT activities, such as online research and educational games, demonstrated higher academic achievement (Adelantado-Renau et al., 2019). Similarly, students who engaged in leisure ICT activities, such as online reading and social networking, exhibited improved academic performance. These findings might suggest that leisure activities involving ICT can contribute to the development of cognitive skills, information processing abilities, and digital literacy, all of which can positively influence academic achievement. By integrating technology into leisure activities, educators and parents can create opportunities for students to enhance their academic skills while enjoying their free time. On the other hand, while using ICT for leisure can enhance cognitive and metacognitive skills, excessive leisure time with ICT, like playing video games, may lead to issues with attention and lower academic performance according to Gentile (2009).

2.6. Self-efficacy in digital competences

In the rapidly evolving digital landscape, digital competences have become essential for individuals to navigate and succeed in various domains of life. Self-efficacy, a psychological construct developed by Bandura (1997), plays a crucial role in determining an individual's ability to effectively utilize and adapt to digital technologies. Self-efficacy refers to an individual's belief in their capability to successfully perform specific tasks or achieve desired outcomes (Zimmerman, 2000). It is a key determinant of human behavior and has been extensively studied across various domains. When applied to digital competences, self-efficacy signifies an individual's confidence in their ability to effectively engage with digital tools, platforms, and information and communication technologies (Katsarou, 2021).

Importance of Self-Efficacy in Digital Competences: Self-efficacy in digital competences is crucial for individuals to thrive in today's technology-driven society. It influences their willingness to adopt and engage with digital technologies, their ability to learn and adapt to new tools and platforms, and their overall digital skill development (Sharma & Saini, 2022). Individuals with higher self-efficacy in digital competences tend to have greater motivation, persistence, and problem-solving abilities when faced with digital challenges (Wang & Chu, 2023).

To foster and enhance self-efficacy in digital competences, it is important to consider different factors. Firstly, providing individuals with hands-on experiences and opportunities for successful digital interactions can boost their confidence (Kauffman & Kauffman, 2017). Setting achievable goals and celebrating progress can also contribute to developing self-efficacy (Sewell & St George, 2000). Additionally, observing others who excel in digital skills, having role models, mentors, and being part of supportive online communities can positively influence an individual's self-efficacy (Mokoena et al., 2021). Positive feedback, constructive encouragement, and support from peers, instructors, and mentors are also crucial in building belief in digital capabilities. Lastly, managing anxiety and stress related to digital technologies through relaxation techniques, stress management, and adopting a growth mindset can help individuals overcome challenges and develop confidence in their digital competences. By addressing these aspects, individuals can improve their self-efficacy in digital skills.

2.7. The non-linear relationship between using ICT and academic achievement

The relationship between ICT usage and academic achievement exhibits a non-linear pattern, indicating that both minimal and excessive use can adversely affect educational outcomes. Recent studies have highlighted that moderate use of ICT resources, such as educational software and online learning platforms, can significantly enhance students' engagement and understanding of complex subjects (Scanlon et al., 2015; Skryabin et al., 2015; Lei & Zhao, 2007). However, excessive use, particularly for non-educational activities, can lead to distractions and reduced academic performance. Students who allocate a substantial amount of time to social media or gaming often experience declines in their academic achievements due to decreased study time and potential cognitive overload (Troll et al., 2021; Giunchiglia et al., 2018; Masood et al., 2020).

Furthermore, the effectiveness of ICT in education heavily relies on the quality of technology integration and the pedagogical strategies employed by educators. Merely incorporating ICT into classrooms without a strategic plan does not guarantee significant improvements in academic performance (Tamim et al., 2011; Ghavifekr & Rosdy, 2015). Effective integration involves not only providing access to technology but also equipping teachers with the necessary skills to integrate these tools into their teaching practices, fostering a more interactive and engaging learning environment. Therefore, while ICT has the potential to substantially enhance academic outcomes, its impact is maximized when used cautiously and supported by effective instructional strategies (Lei et al., 2021; Wang & Wang, 2023).

3. Hypotheses development

This section presents the study's hypotheses, grounded in the theoretical framework and past empirical research. Based on social cognitive and self-regulated learning theories, both linear and non-linear relationships between different forms of ICT use, digital self-efficacy, and students' mathematics and science achievement, as well as the associations between ICT engagement and digital self-efficacy are examined.

Hypothesis 1. Students' ICT use at school, at home, for leisure, and for school-related activities outside the classroom, as well as their digital self-efficacy, are linearly associated with their achievement in mathematics.

Hypothesis 2. Students' ICT use at school, at home, for leisure, and for school-related activities outside the classroom, as well as their digital self-efficacy, are linearly associated with their achievement in science.

Hypothesis 3. ICT use at school, at home, for leisure, and for school-related activities outside the classroom is linearly associated with students' digital self-efficacy.

Hypothesis 4. ICT use at school, at home, for leisure, and for school-related activities outside the classroom, as well as digital self-efficacy, exhibits non-linear relationships with mathematics achievement.

Hypothesis 5. ICT use at school, at home, for leisure, and for school-related activities outside the classroom, as well as digital self-efficacy, exhibits non-linear relationships with science achievement.

Hypothesis 6. The use of ICT at school, at home, for leisure, and for school-related activities outside the classroom exhibits non-linear relationships with students' digital self-efficacy.

4. Conceptual framework and underpinning theory

The conceptual framework of this study is grounded in a synthesis of Social Cognitive Theory, Self-Regulated Learning Theory, Cognitive Load Theory, and Engagement Theory, which together provide a coherent theoretical basis for explaining how ICT engagement influences students' mathematics and science achievement. These complementary perspectives allow the model to account not only for direct associations between ICT use and achievement but also for psychological mechanisms and non-linear effects that characterize technology-enhanced learning environments.

Social Cognitive Theory (Bandura, 1997) provides the primary theoretical foundation for the mediating role of digital self-efficacy in the proposed model. According to this theory, individuals' beliefs about their capabilities strongly influence their behavior, persistence, and performance. In the context of ICT use, students who believe they can effectively use digital tools are more likely to engage productively with technology, apply appropriate learning strategies, and persist when facing academic challenges. Accordingly, the model posits that different forms of ICT engagement, such as ICT use at school, at home, and for school-related activities outside the classroom, shape students' digital self-efficacy, which in turn influences their mathematics and science achievement. This theoretical logic directly supports the hypothesized mediation paths linking ICT engagement to achievement through digital self-efficacy.

Self-Regulated Learning Theory (Zimmerman, 2000) further supports the proposed relationships by emphasizing learners' active role in planning, monitoring, and regulating their learning processes. Purposeful ICT use for academic tasks can facilitate self-regulated learning by providing access to information, feedback, and interactive learning resources. Students with higher digital self-efficacy are better positioned to use ICT strategically, transforming digital engagement into meaningful learning experiences. From this perspective, digital self-efficacy functions as a key mechanism through which ICT engagement contributes to academic outcomes, reinforcing its role as a mediator in the conceptual framework.

Cognitive Load Theory (Sweller, 1988) provides a theoretical rationale for the inclusion of non-linear relationships in the model. While moderate and well-structured ICT use may support learning by enhancing visualization, practice, and engagement, excessive or poorly designed ICT use can impose extraneous cognitive load, diverting attention away from core learning tasks. This is particularly relevant for leisure-oriented ICT use, which may compete with academic activities and reduce effective learning time. Consequently, the framework allows for non-linear relationships between ICT engagement and both digital self-efficacy and academic achievement, reflecting the possibility of diminishing or negative returns at higher levels of ICT use.

The mediating role of digital self-efficacy in this study is theoretically grounded in Social Cognitive Theory and Self-Regulated Learning Theory. According to Social Cognitive Theory (Bandura, 1997), environmental factors such as exposure to and engagement with digital technologies influence academic performance primarily through individuals' beliefs about their capabilities rather than through direct effects alone. In this framework, ICT engagement constitutes the environmental input; digital self-efficacy represents students' perceived competence in using digital tools effectively, and mathematics and science achievement reflect learning outcomes. Students who frequently and purposefully engage with ICT are more likely to develop stronger digital self-efficacy, which in turn enhances their persistence, strategic behavior, and effective use of digital resources in learning tasks. Complementarily, Self-Regulated Learning Theory (Zimmerman, 2000) posits that learners with higher self-efficacy are better able to plan, monitor, and regulate their learning processes when using ICT, thereby translating digital engagement into meaningful academic gains. Without sufficient digital self-efficacy, ICT use may remain superficial or recreational, limiting its instructional value. Accordingly, digital self-efficacy is theoretically positioned as a key psychological mechanism that mediates the relationship between ICT engagement and students' mathematics and science achievement, providing a robust justification for the indirect paths specified in the conceptual model.

Engagement Theory (Kearsley & Shneiderman, 1998) further justifies the differentiation of ICT engagement into distinct modes rather than treating ICT use as a single construct. This perspective emphasizes that learning benefits arise when students are meaningfully and purposefully engaged in tasks that involve interaction, collaboration, and problem solving. In line with this theory, the model distinguishes between ICT use at school, ICT use at home, ICT use for school-related activities outside the classroom, and leisure ICT use, acknowledging that these modes are likely to differ in their educational value and impact on achievement.

Taken together, these theoretical perspectives provide a strong conceptual foundation for the proposed model. They justify the inclusion of multiple ICT engagement dimensions, the mediating role of digital self-efficacy, and the examination of both linear and non-linear relationships with mathematics and science achievement. By explicitly linking theory to each hypothesized path, the conceptual framework moves beyond descriptive associations and offers a theory-driven explanation of how and under what conditions ICT engagement contributes to student learning outcomes within the PISA 2022 context.

The conceptual framework (Fig. 1) for this study aims to explore the complex relationship between students' use of ICT and their academic achievement in mathematics and science.

5. Methods

5.1. Research procedure and sample

The study draws on data from the Türkiye sample of the Program for International Student Assessment (PISA) 2022, which is designed to be nationally representative of 15-year-old students enrolled in formal education in Türkiye. PISA 2022 employed a stratified two-stage probability sampling design. In the first stage, schools were selected with probability proportional to size across explicit and implicit strata, including geographic region and school type. In the second stage, students were randomly selected within sampled schools. This design ensures the inclusion of students from diverse socioeconomic, regional, and institutional backgrounds across Türkiye. As a result, the analytic sample is intended to be reflective of the broader population of 15-year-old students in Türkiye rather than a convenience sample.

This study makes use of secondary data from respondents to the Türkiye Student Questionnaire and the ICT Familiarity Questionnaire from the PISA 2022 database. The OECD used the PISA 2022 evaluation's ICT Familiarity Questionnaire and Student Questionnaire to gather the secondary data set for this study. The 7250 Türkiye pupils (15 years old; 3561 boys and 3689 females) who took part in PISA 2022 are the source of the secondary data.

5.2. Research instruments

5.2.1. ICT available at school (ICTSCH)

In question IC170, which was scaled into the index of availability at school, students were asked to rate how much they use various digital resources at school (e.g., "desktop or laptop computer" and "smartphone (i.e., mobile phone with internet access)". This scale consisted of seven items with six options for responding to each item: "Never or almost never," "About once or twice a month," "About once or twice a week," "Every day or almost every day," "Several times a day," and "This resource is not available to me at school." Before scaling, the response option "This resource is not available to me at school" was recorded as 0, and the other five response options were recorded as 1. For the ICTSCH scale, Cronbach's alpha reliability coefficient is 0.790.

5.2.2. ICT available outside school (ICTHOME)

In question IC171, which was scaled into the index of availability outside school, students were asked to rate how much they use various digital resources at school (e.g., "desktop or laptop computer" and "smartphone (i.e., mobile phone with internet access)"). Each of the six items included in this scale had six response options: Never or almost never," "About once or twice a month," "About once or twice a week," "Every day or almost every day," "Several times a day," and "This resource is not available to me at school". "This resource is not available to me at school" was recorded as 0, and the other five response options were recorded as 1. For the ICTHOME scale, Cronbach's alpha reliability coefficient is 0.771.

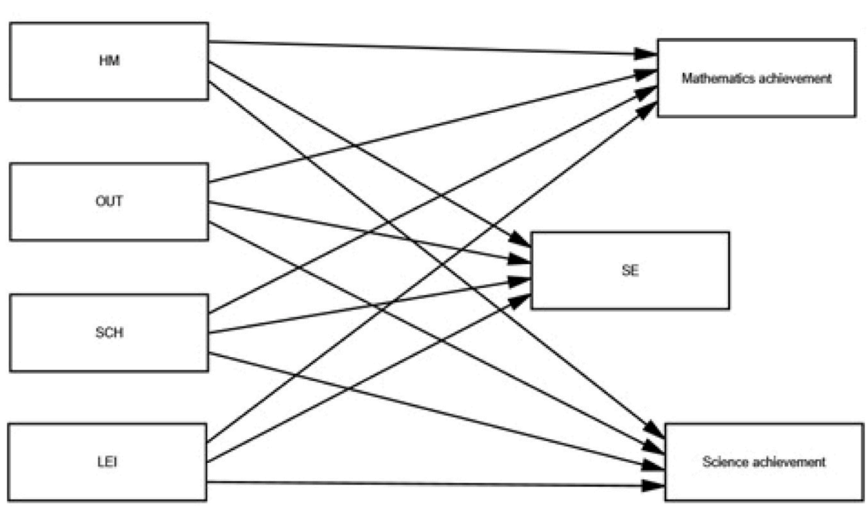

Fig. 1. Conceptual framework. Notes: *HM*: ICT available outside school (ICTHOME), *OUT*: Use of ICT for school activities of the classroom (ICTOUT), *SCH*: ICT available at school (ICTSCH), *LEI*: Leisure activities using ICT (ICTWKDY), SE: Self-efficacy in digital competences (ICTEFFIC).

5.2.3. Leisure activities using ICT (ICTWKDY)

Students were asked to rate how often they did various leisure activities using ICT during a typical weekday (e.g., "play videogames (using my smartphone, a gaming console, or an online platform or apps)"), and look for practical information online in question IC177 was scaled into the index of "Frequency of ICT activity —Weekday." Each of the seven items included in this scale had six response options: "No time at all," "Less than 1 h a day," "Between 1 and 3 h a day," "More than 3 h and up to 5 h a day," "More than 5 h and up to 7 h a day," "more than 7 h a day"). For the ICTWKDY scale, Cronbach's alpha reliability coefficient is 0.861.

5.2.4. Use of ICT for school activities outside of the classroom (ICTOUT)

Students were asked to rate how often they use digital resources for various school-related activities outside of the classroom (e.g., "see my grades or results from specific assignments (e.g., homework or tests)", "communicate with my teacher"). Question IC176 was scaled into the index of "use of ICT for school activities outside the classroom." Each of the eight items included on this scale had five response options: Never or almost never," "about once or twice a year," "about once or twice a month", "about once or twice a week," "every day" or almost every day"). For the ICTOUT scale, Cronbach's alpha reliability coefficient is 0.922.

5.2.5. Self-efficacy in digital competences (ICTEFFIC)

Students were asked to rate how well they can do various tasks using digital resources (e.g., "search for and find relevant information online" and "write or edit text for a school assignment ") in question IC183 was scaled into the index of "self-efficacy in digital competencies." Each of the 14 items included in this scale had four response options: "I cannot do this," " I struggle to do this on my own,", "I can do this with a bit of effort," and "I can easily do this." For the ICTEFFIC scale, the Cronbach's alpha reliability coefficient is 0.93.

5.2.6. Mathematics and science achievement

The PISA 2022 dataset's plausible values for math and science literacy were used to evaluate academic achievement. For each domain (PV1MATH through PV10MATH and PV1SCIE through PV10SCIE), the OECD specifically produced ten plausible values that represented the estimated proficiencies of individual students throughout a distribution of likely scores (OECD, 2023b). These plausible values are multiple imputations taken from Item Response Theory (IRT) models to provide population-level proficiency estimates while taking measurement error into account. They are not test scores in the conventional sense. Each statistical model was performed independently across all ten plausible values in accordance with conventional PISA analytic practice (Rutkowski et al., 2010), and the resulting estimates were averaged to produce reliable conclusions.

6. Data analysis

Since prediction was the main goal of this study, partial least squares structural equation modeling, or PLS-SEM, was used to analyze the data (Hair et al., 2022). Using PLS-SEM was decided in part because of the theoretical model's complexity, resilience, higher-order construct suggestion, and guaranteed higher statistical power. PLS-SEM was chosen over covariance-based SEM (CB-SEM) due to the combination of advanced techniques and intricate bootstrapping methods (Hair et al., 2022). For the PLS_SEM model, SmartPLS 4 (version 4.0.9.4) was utilized (Ringle et al., 2022; Rehman et al., 2025).

First, the validity and reliability of the measurement model were assessed. The statistical characteristics of this approach are as follows: (1) bootstrapping using $n = 10{,}000$ subsamples; (2) convergent validity (reliability of the indicator with the average variance extracted; AVE); (3) discriminant validity using the heterotrait-monotrait ratio (HTMT) and Fornell and Larcker (1981) criteria; and (4) internal consistency using composite reliability and Cronbach's Alpha.

The structural model was evaluated based on the following criteria: out-of-sample prediction (PLSpredict), statistical significance, and the relevance of structural links. Specifically, the effect size (f^2), coefficient of determination (R^2), predictive relevance (Q^2), and relevant prediction errors of the model were tested.

7. Results

The research questions were addressed by using measures of students' ICT use in general at school, for leisure outside of school, interest in ICT, perceived ICT competency, and math and science achievement from PISA 2022. The study's ICT constructs were chosen for their theoretical significance, psychometric strength, and alignment with previous studies (Courtney et al., 2022; Zhu & Li, 2022). Digital self-efficacy and ICT use for school-related tasks outside of the classroom, at home, and in leisure were included. These dimensions showed conceptual clarity and excellent internal consistency (Cronbach's $\alpha > 0.80$).

Preliminary analyses were carried out to assess the properties of important study variables before doing hypothesis testing using PLS-SEM. For every construct, including ICT use at school (ICTSCH), ICT use outside of school (ICTHOME), ICT use for leisure (ICTWKDY), ICT use for school-related tasks outside of class (ICTOUT), and self-efficacy in digital competencies (ICTEFFIC), Table 1 displays descriptive statistics (mean, standard deviation, skewness, and kurtosis).

The ICTHOME scale has the highest mean score ($M = 3.33$, $SD = 1.17$), while gender has the lowest ($M = 1.51$, $SD = 0.50$). The univariate normality of the data was assessed by examining skewness and kurtosis indices, which ranged from −0.52–0.45 and −0.79–0.39, respectively, across different scales. According to Kline's (2016) guidelines, stating that skewness should be less than the absolute value of 3 and kurtosis should be less than the absolute value of 10, the data in this study were considered normal and suitable for further analysis.

The relationship between the constructs was investigated using the Pearson product-moment correlation coefficient. Preliminary analyses were conducted to ensure no violations of the assumptions of normality, linearity, and homoscedasticity occurred. Several significant correlations emerged. For instance, as shown in Table 2, there was a moderately positive correlation between ICTEFFIC and ICTOUT ($r = .33$, $p < .001$), indicating that students' self-efficacy in digital skills is correlated with their level of access to ICT outside class. Both MATH ($r = -.15$, $p < .001$) and SCIENCE ($r = -.13$, $p < .001$) achievements were negatively correlated with ICTWKDY, supporting earlier findings that excessive ICT use for leisure may impair academic performance.

The inclusion of self-efficacy in digital competencies as a potential mediator was supported by the small correlations it showed with both math achievement ($r = .15$, $p < .001$) and science achievement ($r = .17$, $p < .001$). The inclusion of the research variables in the structural equation model is justified by these descriptive and correlational results, which also support the validity of the proposed paths. Gender was included as a covariate in the analysis. From Table 2, gender positively correlated with ICTWKDY ($r = .14$, $p < .01$), ICTSCH ($r = .07$, $p < .01$), ICTHOME ($r = .08$, $p < .01$), and mathematics achievement ($r = .03$, $p < .01$). Gender was negatively associated with ICTOUT ($r = -.18$, $p < 0.01$) and science achievement ($r = -.03$, $p < .05$). However, gender

Table 1
Descriptive statistics.

Variable	Mean	SD	Skewness	Kurtosis
Gender	1.51	0.50	-0.04	-2.00
ICT use at school (ICTSCH)	2.89	1.27	0.48	-0.54
ICT use outside school (ICTHOME)	3.33	1.17	-0.05	-0.58
ICT use for leisure (ICTWKDY)	2.75	1.06	0.88	0.94
ICT use outside class (ICTOUT)	3.29	1.01	-0.45	-0.21
Digital self-efficacy (ICTTEFFIC)	3.21	0.74	-0.85	1.17

Gender was coded as dummy variable (female = 1, male = 2)

Table 2
Pearson correlations among key variables.

Variable	1	2	3	4	5	6	7	8
1. Gender	1							
2. ICT use at school (ICTSCH)	.07**	1						
3. ICT Use Outside School (ICTHOME)	.08**	.51**	1					
4. ICT Use for Leisure (ICTWKDY)	.14**	.25**	.30**	1				
5. ICT Use Outside Class (ICTOUT)	-.18**	.20**	.24**	.15**	1			
6. Digital Self-Efficacy (ICTEFFIC)	.01	.18**	.26**	.27**	.33**	1		
7. Math Achievement (PV_MATH)	.03**	.02	.16**	-.15**	.10**	.15**	1	
8. Science Achievement (PV_SCI)	-.03*	.01	.17**	-.13**	.14**	.17**	.94**	1

Note: ***p** < .01; *p < .05; Gender was coded as dummy variable (female = 1, male = 2)

was not significantly correlated with ICTEFFIC (r = .01, p = .96).

The model was estimated by using the PLS-SEM technique from the SmartPLS 4 program (version 4.0.9.5) (Ringle et al., 2022) with default initial weights and a maximum of 3000 iterations. As advised by Ringle et al. (2022), bootstrapping which is a nonparametric method that runs 5000 samples, was used to ascertain the statistical significance of the PLS-SEM results.

7.1. Evaluation of the measurement model

Indicator loadings were used to examine the reflectively expressed structures. Indicator loadings larger than 0.708 are advised by Hair et al. (2019) since they show that the concept explains more than 50 % of the indicator's variance, providing adequate item dependability. Table 3 shows that the outer loading value of each indicator is more than 0.708.

The internal consistency and reliability were assessed using Jöreskog's (1971) composite reliability (ρC). Values between 0.70 and 0.90 are categorized by Hair et al. (2019) as "satisfactory to good," whereas reliability levels between 0.60 and 0.70 are seen as "acceptable in exploratory research." Higher and more concerning values of 0.95 imply that the items are redundant, which lowers the construct validity.

Although PLS-SEM does not provide a global goodness-of-fit index equivalent to CB-SEM, the model’s quality was evaluated by numerous complementary criteria. A successful model fit was shown by the Standardized Root Mean Square Residual (SRMR), which was calculated and produced a value of 0.06, below the cautious criterion of 0.08 (Henseler et al., 2016). For every construct, the Average Variance Extracted (AVE) was greater than 0.50, indicating convergent validity. Multicollinearity issues were eliminated because all the variance inflation factors (VIFs) were less than 3.3. All these values support the fact that the model's specification is adequate.

Even though Cronbach's alpha employs equivalent criteria, it performs worse than composite reliability as a measure of internal consistency dependability (Hair et al., 2019). Convergent validity for each idea was examined in the following phase of the reflective measurement method. The convergent validity of the construct was assessed using the average variance extracted, or AVE. A construct is considered acceptable, per Hair et al. (2019), if its average variance exponent (AVE) is 0.50 or greater and it explains at least half of the variation among its constituent elements.

Table 3
Evaluation of the measurement model.

Item	ICTSCH	ICTHOME	ICTWKDY	ICTOUT	ICTEFFIC	alpha	CR	AVE	VIF
ICTSCH						.790	.838	.597	
SCH1	0.795								1.411
SCH2	0.738								1.628
SCH3	0.701								1.652
SCH4	0.849								1.642
ICTHOME						.771	.800	.685	
HM1		0.741							1.401
HM2		0.873							1.715
HM4		0.862							1.836
ICTWKDY						.861	.870	.705	
LEI4			0.834						2.092
LEI5			0.804						1.760
LEI6			0.851						2.243
LEI7			0.867						2.106
ICTOUT						.922	.934	.645	
OUT1				0.797					2.229
OUT2				0.840					2.739
OUT3				0.858					2.920
OUT4				0.846					2.877
OUT5				0.751					2.113
OUT6				0.725					2.009
OUT7				0.783					2.469
OUT8				0.813					2.566
ICTEFFIC						.894	.898	.654	
SE1					0.802				2.438
SE2					0.847				2.819
SE3					0.832				2.363
SE4					0.819				2.412
SE5					0.785				2.241
SE6					0.762				1.870

Notes: CR: composite reliability; AVE: average variance extracted; VIF: variance inflation factors.
ICTSCH: ICT available at school; ICTHOME: ICT available outside school; ICTWKDY: Leisure activities using ICT; ICTOUT: Use of ICT for school activities in the classroom; ICTEFFIC: Self-efficacy in digital competences.

Reliability coefficients, composite reliability, and Cronbach's alpha values of more than 0.7 for each construct in Table 3 demonstrate strong internal consistency. Discriminant validity, which measures how much a construct varies empirically from other structural model elements, was evaluated using two techniques. Fornell and Larcker (1981) advised comparing each construct's AVE to the square inter-construct correlation of that construct and each other reflectively rated construct in the structural model. They also proposed that the shared variance of no model construct should be higher than the AVE.

The second criterion, heterotrait-monotrait (HTMT) ratio of the correlations, was developed by Henseler et al. (2015) to evaluate discriminant validity. According to Henseler et al. (2015), the geometric mean of average correlations for items measuring the same construct is divided by the average correlation value for distinct constructs to determine the HTMT.

Henseler et al. (2015) suggest a threshold value of 0.90 for structural models with comparatively similar structures; they suggest a value of 0.85 for structural models with fundamentally dissimilar constructs. Consequently, when HTMT values are higher than 0.85 or 0.90, discriminant validity issues occur. Tables 4 and 5 demonstrate that sufficient discriminant validity has been attained for the current study based on the evaluation results for the HTMT and Fornell-Larcker criteria.

7.2. Assessing the structural model

7.2.1. Linear relationship

Table 6 shows that ICTSCH (β = −0.055, t = 4.304), ICTHOME (β = 0.207, t = 22.58), ICTWKDY (β = −0.267, t = 25.03), and ICTEFFIC (β = 0.216, t = 18.538) have a statistically significant linear relationship with mathematics achievement. Thus, H1 was supported, except for ICTOUT (β = 0.012, t = 0.98), which did not have a statistically significant linear relationship with mathematics achievement. Similarly, ICTSCH (β = −0.051, t = 4.034), ICTHOME (β = 0.275, t = 22.674), ICTOUT (β = 0.040, t = 3.384), ICTWKDY (β = −0.265, t = 24.456), and ICTEFFIC (β = 0.227, t = 19.313) have a statistically significant linear relationship with science achievement. Hence, H2 was supported. In addition, ICTHOME (β = 0.207, t = 15.128), ICTOUT (β = 0.318, t = 25.101), and ICTWKDY (β = 0.053, t = 4.319) were significantly and linearly related to ICTEFFIC. Therefore, H3 was supported, except for ICTSCH (β = 0.001, t = 0.058), which did not have a statistically significant linear relationship with ICTEFFIC.

7.2.2. Non-linear relationship

Table 7 shows that ICTHOME (β = - 0.022, t = 2.292) and ICTOUT (β = −0.060, t = 6.927) have a significant non-linear relationship with mathematics achievement. Thus, H4 is supported, except for ICTEFFIC (β = −0.008, t = 0.929), which did not have a significant non-linear relationship with mathematics achievement. The findings indicate that there were significant non-linear relationships between ICTHOME (β = −0.031, t = 2.955), ICTSCH (β = −0.043, t = 4.151), ICTWKDY (β = 0.025, t = 2.893), and ICTEFFIC (β = −0.027, t = 3.331) and science achievement. Thus, H5 is supported.

Similarly, there was a significant non-linear relationship between ICTHOME (β = −0.051, t = 4.212), ICTOUT (β = −0.032, t = 2.919), ICTSCH (β = −0.025, t = 2.225), and ICTEFFIC. Hence, H6 is supported, except for ICTWKDY (β = −0.001, t = 0.063), which did not have a significant linear relationship with ICTEFFIC.

Table 4
Application of the Fornell-Lacker criterion to assess discriminant validity.

Dimensions	ICTEFFIC	ICTHOME	ICTOUT	ICTSCH	ICTWKDY
ICTEFFIC	**(0.805)**				
ICTHOME	0.297	**(0.827)**			
ICTOUT	0.380	0.263	**(0.803)**		
ICTSCH	0.168	0.435	0.208	**(0.773)**	
ICTWKDY	0.125	0.128	0.141	0.207	**(0.839)**

Note: The diagonal's bolded values correspond to AVE's square root.

Table 5
Making use of the HTMT criterion to assess discriminant validity.

	ICTEFFIC	ICTHOME	ICTOUT	ICTSCH	ICTWKDY
ICTEFFIC					
ICTHOME	0.353				
ICTOUT	0.413	0.303			
ICTSCH	0.184	0.548	0.232		
ICTWKDY	0.148	0.168	0.173	0.256	

7.2.3. Mediation analysis of self-efficacy in digital competence

The direct, indirect, and total effects were investigated using the bootstrapping approach with 5000 samples to evaluate the mediating impact of self-efficacy in digital competences (ICTEFFIC) between ICT use aspects and student accomplishment (Preacher & Hayes, 2008).

The results indicated that, for Example:

1. ICTOUT → ICTEFFIC → Math Achievement: Indirect effect = 0.068 (t = 6.54, $p < .001$)
2. ICTHOME → ICTEFFIC → Science Achievement: Indirect effect = 0.063 (t = 5.91, $p < .001$)

These results indicate partial mediation, where ICT use outside school and school-related tasks enhances academic outcomes in math and science through increased confidence in digital abilities. Table 8 reports on the mediation pathways to further substantiate the claims.

8. Discussion

The findings make three interrelated contributions to understanding the relationship between ICT engagement and academic achievement. First, different modes of ICT use are associated with distinct achievement patterns in mathematics and science. Academically oriented ICT engagement, particularly use related to school tasks outside the classroom, is positively associated with achievement, whereas leisure-oriented ICT use shows negative associations. These modality-based differences are consistent with prior meta-analytic and large-scale studies emphasizing that the educational value of technology depends on its pedagogical purpose rather than mere access or frequency of use (Tamim et al., 2011; Lei & Zhao, 2007; Remón et al., 2017). Similar patterns have been reported in recent empirical work showing that structured, school-related ICT use is more strongly linked to learning outcomes than unstructured or entertainment-focused digital activities (Sari, 2022). In contrast, leisure-oriented screen time, including social media and gaming, tends to correlate negatively with achievement, endorsing prior findings linking intensive recreational ICT use to increased stress, distraction, and weaker academic performance (Masood et al., 2020; Bulut & Cutumisu, 2018).

Second, digital self-efficacy emerges as a prominent mediating mechanism. Consistent with Social Cognitive Theory and self-regulated learning frameworks, students who report greater confidence in their digital skills derive stronger academic benefits from school-related ICT engagement (Bandura, 1997; Zimmerman, 2000). As shown in Table 8, a meaningful portion of the association between academically oriented ICT use and both mathematics and science achievement operates indirectly through digital self-efficacy, whereas corresponding indirect effects for leisure-oriented ICT use are weaker or non-significant. These findings align with prior research suggesting that when ICT engagement supports students' perceptions of competence and autonomy, it is more likely to foster motivation and engagement (Lei & Zhao, 2007; Ryan & Deci, 2017; Chen et al., 2024).

Third, the association between ICT intensity and achievement is not

Table 6
Hypothesis testing of linear effects.

	Path	Beta	t-value	CI	Supported	f^2	Q^2
H1	*ICTSCH*→ Math achievement	-0.055	4.304***	[-.078, −.029]	Yes	0.003	0.032
	ICTHOME→ Math achievement	0.207	22.58***	[0.25, 0.29]	Yes	0.051	
	ICTWKDY→ Math achievement	-0.267	25.03***	[-0.29, −0.25]	Yes	0.085	
	ICTOUT → Math achievement	0.012	0.98 ns	[-0.012, 0.035]	No	0.000	
	ICTEFFIC→*Math* achievement	0.216	18.538***	[0.19, 0.24]	Yes	0.026	0.127
H2	ICTSCH→ Science achievement	-0.051	4.034***	[-0.76, −0.026]	Yes	0.001	
	ICTHOME→ Science achievement	0.275	22.674***	[0.250, 0.298]	Yes	0.051	
	ICTOUT→ Science achievement	0.040	3.384***	[0.016, 0.063]	Yes	0.001	
	ICTWKDY→ Science achievement	-0.265	24.456***	[-0.286, −0.244]	Yes	0.064	0.119
	ICTEFFIC→Science achievement	0.227	19.313***	[0.204, 0.251]	Yes	0.024	
H3	ICTSCH→ICTEFFIC	0.001	0.058 ns	[-0.023, 0.025]	No	0.000	
	ICTHOME→ICTEFFIC	0.207	15.128***	[0.180, 0.234]	Yes	0.023	
	ICTWKDY→ICTEFFIC	0.053	4.319***	[0.029, 0.078]	Yes	0.002	
	ICTOUT→ICTEFFIC	0.318	25.101***	[0.293, 0.343]	Yes	0.087	

**$p < .01$; *$p < .05$; ***$p < .001$

Table 7
Hypothesis testing of nonlinear effects.

	Path	Beta	t-value	CI	Supported	f^2	Q^2
H4	ICTHOME→ Math achievement	-0.022	2.292**	[-.041, −.003]	Yes	0.001	0.032
	ICTOUT → Math achievement	-0.060	6.927***	[-0.078, −0.044]	Yes	0.006	
	ICTEFFIC→ Math achievement	-0.008	0.929 ns	[-0.023, 0.008]	No	0.000	
H5	ICTHOME → Science achievement	-0.031	2.955***	[-0.051, −0.010]	Yes	0.001	
	ICTSCH→*Science* achievement	-0.043	4.151***	[-0.063, −0.023]	Yes	0.002	0.127
	ICTWKDY→ Science achievement	0.025	2.893***	[0.008, 0.042]	Yes	0.001	
	ICTEFFIC→ Science achievement	-0.027	3.331***	[-0.044, −0.012]	Yes	0.002	
H6	ICTHOME→ ICTEFFIC	-0.051	4.212***	[-0.075, −0.027]	Yes	0.003	
	ICTOUT→ ICTEFFIC	-0.032	2.919***	[-0.054, −0.010]	Yes	0.002	0.119
	ICTSCH→ICTEFFIC	-0.025	2.225***	[-0.047, −0.002]	Yes	0.001	
	ICTWKDY→ICTEFFIC	-0.001	0.063 ns	[-0.019, 0.018]	No	0.000	

**$p < .01$; *$p < .05$; ***$p < .001$

Table 8
Mediation pathways: ICT Use → ICTEFFIC → mathematics and science achievement.

Path	Direct effect	Indirect effect	Total effect	t-value (indirect)	95 % CI (indirect)	Mediation type
ICTSCH→ ICTEFFIC →*Math*	-.055	.000	-.055	.058	[-.010,.011]	None
ICTHOME→ ICTEFFIC →*Math*	.207	.045	.252	4.94***	[.028,.063]	Partial
ICTOUT→ I CTEFFIC →*Math*	.012	.069	.081	6.54***	[.048,.091]	Full
ICTWKDY→ I CTEFFIC →*Math*	-.267	.014	-.253	2.82**	[.004,.024]	Partial
ICTSCH→ ICTEFFIC →*Science*	-.051	.000	-.051	.07	[-.011,.012]	None
ICTHOME→ ICTEFFIC →*Science*	.275	.063	.338	5.91***	[.042,.084]	Partial
ICTOUT→ ICTEFFIC →*Science*	.040	.072	.112	6.78***	[.050,.095]	Partial
ICTWKDY→ ICTEFFIC→*Science*	-.265	.017	-.248	3.01**	[.006,.029]	Partial

Note: Bootstrapped standard errors based on 5000 resamples. **$p < .01$, ***$p < .001$

purely linear. The statistically significant quadratic terms reported in Table 7 indicate that the marginal association between ICT use and achievement varies across the observed range of use. In substantive terms, this curvature suggests that additional ICT engagement is not associated with uniform gains at all levels of exposure; rather, the strength and, in some cases, the direction of the association change as ICT use increases. This pattern is consistent with theoretical expectations derived from Cognitive Load Theory, which posits that excessive or poorly structured digital engagement may overwhelm learners' cognitive resources and reduce learning benefits (Sweller, 1988).

Importantly, the presence of statistically significant curvature does not by itself identify a precise turning point or a single "optimal" level of ICT use. The location and magnitude of any peak in the association depend on the scaling and measurement of the ICT indicators, the observed distribution of use, and the specific model specification employed. For this reason, the findings are interpreted cautiously within the observed range of ICT engagement among students in Türkiye, academically oriented ICT use is associated with positive outcomes at lower levels of intensity, while these associations tend to weaken and, in some cases, reverse at higher levels of use. This interpretation reflects the direction and practical meaning of the estimated coefficients without overstating what can be inferred from the quadratic specification.

A related consideration concerns functional form. The quadratic term provides a parsimonious representation of non-linearity, but it is not the only way the relationship between ICT use and achievement may deviate from linearity. Alternative specifications, such as threshold models, flexible non-linear approaches, or categorical distinctions between low, medium, and high levels of use, may capture different aspects of this relationship and could yield somewhat different substantive interpretations. Accordingly, the present findings are framed as evidence of diminishing or changing returns to ICT engagement within the sample, rather than as confirmation that a single quadratic function fully characterizes the underlying exposure–outcome relationship. This cautious interpretation is consistent with recent PISA-based studies that report non-linear patterns of ICT engagement without converging on a single universal threshold (Agasisti et al., 2020; Zhu & Li, 2022).

Taken together, these results highlight the complexity of ICT–achievement relationships in an emerging-economy context. The

Türkiye-specific evidence suggests that access to digital resources interacts with students' readiness to use them effectively, helping to explain why unguided ICT engagement often shows weak or inconsistent associations with learning outcomes (Bulut & Cutumisu, 2018). Conversely, ICT activities that are structured, interactive, and aligned with instructional goals such as simulations, virtual laboratories, and collaborative digital tasks are more consistent with principles of Engagement Theory and are more likely to be associated with positive academic outcomes (Kearsley & Shneiderman, 1998). Overall, the findings reinforce the conclusion that ICT can support learning when its use is purposeful and when students possess the digital self-efficacy needed to engage productively with technology, while also highlighting the importance of interpreting non-linear effects with precision and restraint.

9. Implications

9.1. Theoretical implications

The results contribute to several learning theories by clarifying the conditions under which ICT promotes achievement. Social Cognitive Theory (Bandura, 1997) is supported and extended. Digital self-efficacy emerged as a critical mechanism linking ICT engagement to outcomes. That is, students' beliefs about their ICT capabilities strongly influenced their learning results, confirming Zimmerman's (2000) principle that self-efficacy is an "essential motive to learn." This theory is refined by situating it in the digital domain. In effect, it is not merely access to technology that matters, but the learner's confidence and agency in using it (Hatlevik et al., 2018). Likewise, Self-Regulated Learning Theory (Zimmerman, 2000) is upheld. ICT tools can scaffold planning, monitoring, and strategic study only when learners actively engage. The mediation findings imply that ICT use for schoolwork enhances self-regulation indirectly through increased confidence, suggesting that educators should combine technology with scaffolds that build student agency.

The differentiated effects of ICT use also confirm Engagement Theory (Kearsley & Shneiderman, 1998). It is shown that only meaningful ICT tasks that immerse students in interactive, cognitively engaging activities were positively related to achievement. Merely spending time on digital devices without purpose did not foster deep learning, reinforcing the theory's emphasis on active, collaborative learning experiences. In parallel, the observed non-linear pattern validates Cognitive Load Theory (Sweller, 1988). The evidence indicates an optimal amount of ICT. Moderate, well-designed use reduced extraneous load and enhanced learning, while excessive use likely imposed cognitive overload and hindered it. No single theory fully anticipated this conditional effect; by integrating these frameworks, this study offers a novel synthesis. It is suggested that effective ICT integration requires balancing engagement and self-efficacy against cognitive constraints (Bandura, 1997; Sweller, 1988). In short, technology's promise depends on designing tasks that engage students and on cultivating learners' confidence, an insight that refines existing educational psychology models for the digital age.

9.2. Practical implications

The evidence points to clear, actionable strategies for educators and policymakers in Türkiye and similar contexts to employ ICT effectively:

- Prioritize purposeful ICT over mere access: Simply expanding device availability is not enough (Bulut & Cutumisu, 2018). Instead, national strategies should emphasize teacher training and curricular integration that guide students toward effective use. For example, teacher professional development can focus on project-based, technology-rich learning (Onah et al., 2021; Corchuelo-Fernández et al., 2022) so that in-class and homework ICT use directly supports learning objectives. This aligns with OECD (2023) advice to leverage digital devices for specific learning tasks such as up to one hour of guided online learning yielded a measurable score gain while avoiding unguided screen time.
- Build students' digital literacy and self-efficacy: Because confidence in using technology mediated achievement, curricula should explicitly teach ICT skills and self-regulated learning strategies. Embedding ICT projects (coding, multimedia presentations, virtual labs) can help students develop competence and autonomy (Corchuelo-Fernández et al., 2022; Livingstone et al., 2017). Education ministries might adopt digital literacy frameworks and assessments, ensuring that device rollouts are paired with skill-building interventions. For instance, regular use of digital learning platforms with feedback can incrementally boost students' self-efficacy and motivation (Onah et al., 2021; Chen et al., 2024).
- Leverage blended and asynchronous learning but monitor recreational use: Schools should expand technology-supported instruction outside traditional class time to reinforce math and science content (Ogundipe & Akinyemi, 2025; Chen et al., 2025). The positive finding for out-of-school academic ICT use suggests benefits from interactive e-homework tools and educational games (Sun & Hsieh, 2018; Sari, 2022). At the same time, educators and parents should be vigilant about excessive leisure screen time. For example, directing students toward curriculum-aligned educational apps can turn some "recreational" screen hours into learning opportunities. National guidelines or school policies might promote at most moderate daily screen time for non-instructional use, consistent with research cautioning that too much unfocused digital media can distract (OECD, 2023c; Masood et al., 2020).
- Focus on equity and community support: The results show that technology benefits amplify existing skill gaps. Digitally confident students gained more, so disadvantaged students risk falling further behind. In Türkiye and other emerging economies, this implies that infrastructure investments such as broadband, devices must be accompanied by inclusive programs. Efforts could include community digital literacy workshops, family-targeted ICT training, and free internet access in underserved areas (Livingstone et al., 2017; Budiarto et al., 2024). By combining access with capacity-building, policymakers can ensure that no student is left out of the digital learning revolution. Development agencies and NGOs should use these findings to target interventions, for example, by sponsoring after-school programs that help lower-performing students build technology skills so that ICT becomes a tool for closing rather than widening achievement gaps.

10. Limitations and future research recommendations

The study has several limitations that should be acknowledged. First, the cross-sectional nature of PISA 2022 data restricts causal interpretation. Although SEM reveals associations among ICT engagement, digital self-efficacy, and achievement, these relationships remain correlational and may be influenced by unobserved factors. Future research should employ longitudinal or experimental designs to more robustly examine causal pathways, particularly the role of ICT interventions in strengthening digital self-efficacy. Second, the analysis did not include school- or classroom-level variables such as teacher quality, pedagogical practices, or ICT infrastructure. As these contextual factors may shape how technology is used for learning, future studies should apply multilevel SEM to account for hierarchical influences and explore interactions between individual ICT use and school context (Wang & Wang, 2023). Incorporating national or regional policy indicators, such as the implementation of Türkiye's FATIH initiative, would further clarify policy-level moderation effects.

Third, focusing exclusively on Türkiye limits the generalizability of the findings. Given evidence that ICT-achievement relationships differ across countries, future research should replicate the model using PISA 2022 data from multiple contexts. Cross-national and multigroup SEM analyses would help determine which relationships are context-specific and which are more universal (Courtney et al., 2022; Li & Zhu, 2023). Finally, ICT use was measured through self-reports, and effect sizes were small to moderate. Future research could strengthen validity by combining self-reported data with objective usage indicators or experimental measures. Larger samples or targeted analyses of disadvantaged subgroups may also help identify stronger or more differentiated effects.

Addressing these limitations will strengthen the evidence base. In particular, adopting multilevel analytic approaches such as multilevel SEM as in Wang and Wang (2023) and cross-national PISA comparisons can test whether the conclusions about optimal ICT use and self-efficacy hold broadly. Such work will clarify how policy and classroom factors shape the ICT–achievement link across contexts.

11. Conclusion

This study provides robust evidence that in the digital age, quality of use outweighs quantity of use when it comes to ICT engagement and student achievement in mathematics and science. The results affirm that ICT has the potential to support deep learning only when students meaningfully engage with technology in ways that align with pedagogical goals and when learners are equipped with the requisite digital skills and self-efficacy to use these tools effectively. Recent evidence further supports the central role of digital literacy and self-efficacy in enhancing students' engagement and academic outcomes. Students' digital literacy and positive attitudes toward technology have been shown to significantly boost self-efficacy, which in turn improves online learning engagement and academic performance (Getenet et al., 2024). This confirms the findings that students with greater digital confidence realize more substantial learning gains from ICT use (Getenet et al., 2024). Moreover, comparative research across economic contexts indicates that the interplay between digital resource utilization, engagement, and self-efficacy varies across countries but consistently predicts achievement when use is purposeful (Joshi et al., 2025).

Importantly, this study challenges overly optimistic narratives that equate more technology with better learning outcomes. This aligns with recent studies showing that unmediated ICT use or teacher-centered technology integration may not yield the desired academic benefits unless educators purposefully integrate ICT into instruction in ways that promote conceptual understanding and higher-order thinking (Pérez Echeverría et al., 2025). Thus, the present evidence emphasizes that students' technology use should be embedded within carefully structured pedagogical practices rather than being an end in itself. Technology simply increases screen time without support, strategic alignment with curriculum, or development of students' competencies risks cognitive overload, distraction, and even negative effects on achievement.

For educators and policymakers, the implications of this study are both clear and compelling. First, investments in hardware and infrastructure must be matched by investments in teacher capacity building and professional development that enable instructors to design and implement ICT-rich learning experiences effectively. Recent PISA 2022 policy analyses reveal that teachers who receive preparation in ICT skills both during initial training and through ongoing professional development are more likely to use digital resources in ways that support student learning (OECD, 2025). Enhancing teacher readiness for digital innovation will therefore be critical to translating technology investments into improved outcomes. Second, educational systems should integrate student-centered digital literacy curricula that explicitly build learners' confidence and strategic competence with technology. As frontline research demonstrates, greater digital literacy not only strengthens self-efficacy but also leads to greater engagement with digital tools and ultimately to better academic achievement (Getenet et al., 2024; Joshi et al., 2025). Providing structured opportunities for students to use technology for problem-solving, collaboration, and self-directed learning can amplify these benefits and help close achievement gaps. Third, digital policy frameworks should emphasize balanced and equitable access, recognizing that ICT engagement alone will not improve outcomes unless accompanied by instructional coherence and support for all learners. In contexts where access disparities persist, targeted interventions such as community-based digital literacy programs and support for disadvantaged students can help ensure that the benefits of ICT extend across socio-economic levels.

Overall, this study strengthens the case for nuanced and evidence-based ICT integration in education. By emphasizing purposeful use, teacher preparedness, and learner readiness, schools in Türkiye and around the world can employ ICT as a catalyst for 21st-century learning and ensure that students build both digital competencies and strong foundations in STEM subjects. The message for policy and practice is explicit. Invest not only in technology but in the pedagogical and human capital necessary to make it work for all students.

Statements and Declarations

- The data used in this study are available to the public under OECD library: https://www.oecd-ilibrary.org/education/pisa_19963777. Example from: https://www.oecd-ilibrary.org/education/pisa_19963777.
- There are no relevant financial or non-financial competing interests to report.
- No potential competing interest was reported by the authors.
- The authors declare no potential conflicts of interest with respect to the research and authorship pertaining to this article.
- During the preparation of this work the author(s) used Grammarly and Quillbot to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Ernest Afari: Writing – original draft, Methodology, Formal analysis, Data curation. **Funda Örnek:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

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